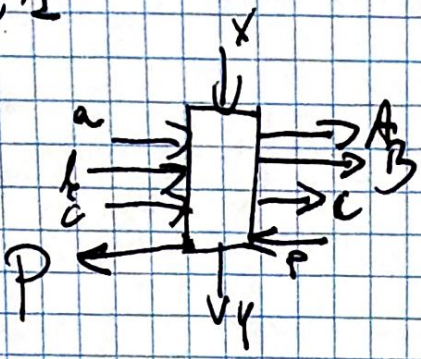
Zad 1

Schematów nie rysowałem



a b
0 0 brak symboli z poszukiwanej sekwencji
0 1 pierwszy bit z sekwencji został zdekodowany
1 0 drugi
1 1 trzeci

P = 1 wystąpi chociaż jedna jedynka

C = 1 gdy wystąpi druga jedynka

c a b x	C A B
0 0 0 0	0 0 0
0 0 0 1	0 0 1
0 0 1 0	0 1 0
0 0 1 1	1 0 1
0 1 0 0	0 1 1
0 1 0 1	1 0 1
0 1 1 0	1 0 0
0 1 1 1	1 0 1
1 0 0 0	1 0 0
1 0 0 1	1 0 1
1 0 1 0	1 1 0
1 0 1 1	1 0 1
1 1 0 0	1 1 1
1 1 0 1	1 0 1
1 1 1 0	1 0 0
1 1 1 1	1 0 1

ca \ bx	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	1	1
10	1	1	1	1

$C = c + bx + ax \cdot ab$

$\overline{C} = \overline{c} \cdot \overline{bx} \cdot \overline{ax} \cdot \overline{ab}$

ca \ bx	00	01	11	10
00	0	0	0	1
01	1	0	0	0
11	1	0	0	0
10	0	0	0	1

$A = a\bar{b}\bar{x} + \bar{a}b\bar{x}$

$Y = cabx + abpx$

$= abx(c+p)$

$= \overline{abcx} \cdot \overline{abpx}$

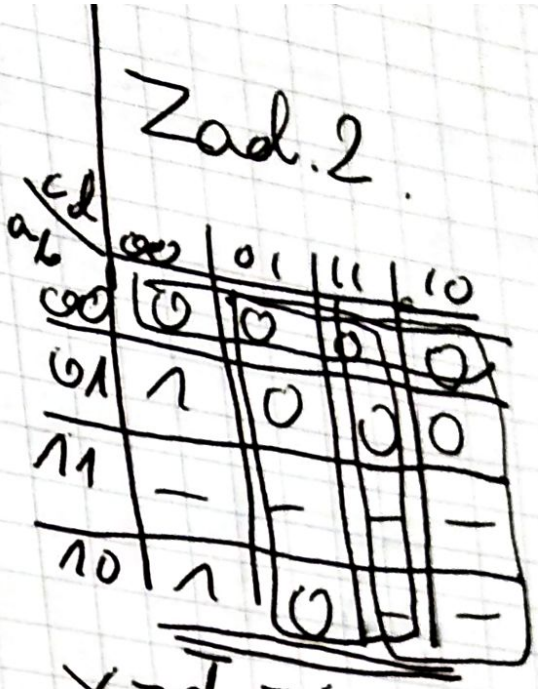
ca \ bx	00	01	11	10
00	0	1	1	0
01	1	1	1	0
11	1	1	1	0
10	0	1	1	0

$B = X + a\bar{b}$

Zad. 2

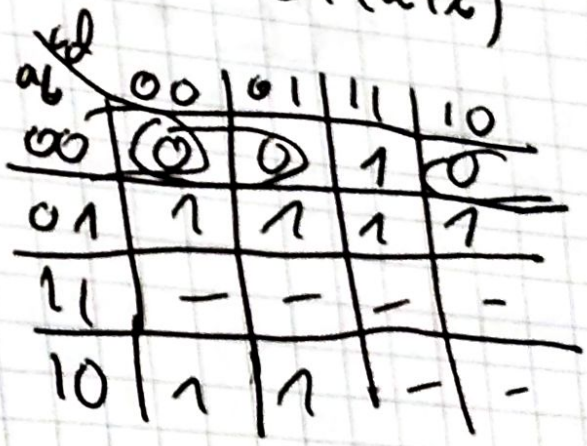
BCD

a b c d	X Y Z
0 0 0 0	0 0 1
0 0 0 1	0 0 1
0 0 1 0	0 0 1
0 0 1 1	0 1 1
0 1 0 0	1 1 1
0 1 0 1	0 1 1
0 1 1 0	0 1 1
0 1 1 1	0 1 0
1 0 0 0	1 1 0
1 0 0 1	0 1 0
1 0 1 0	- - -
1 0 1 1	- - -
1 1 0 0	- - -
1 1 0 1	- - -
1 1 1 0	- - -
1 1 1 1	- - -



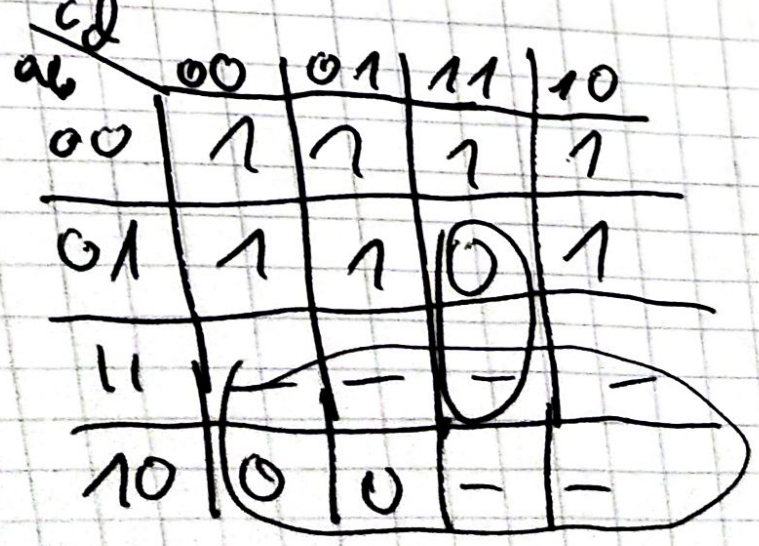
$$X = \overline{d \cdot \bar{c} \cdot (a+b)}$$

$$X = d + c + \overline{(a+b)}$$



$$Y = \overline{(a+b+c)(a+b+d)}$$

$$Y = \overline{a+b+c} + \overline{a+b+d}$$

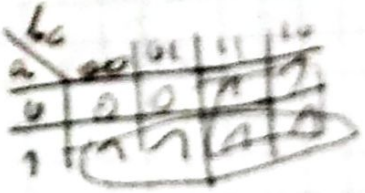


$$Z = \overline{\bar{a}\left(\bar{b}+\bar{c}+\bar{d}\right)}$$

$$Z = a + b + c + d$$

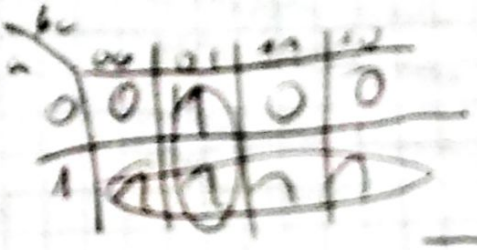
Zad 3.

a b c	X Y
0 0 0	0 0
0 0 1	0 1
0 1 0	1 0
0 1 1	1 0
1 0 0	1 1
1 0 1	1 1
1 1 0	1 1
1 1 1	1 1

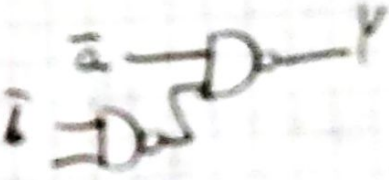


$X = a + b = \overline{\bar{a} \cdot \bar{b}}$

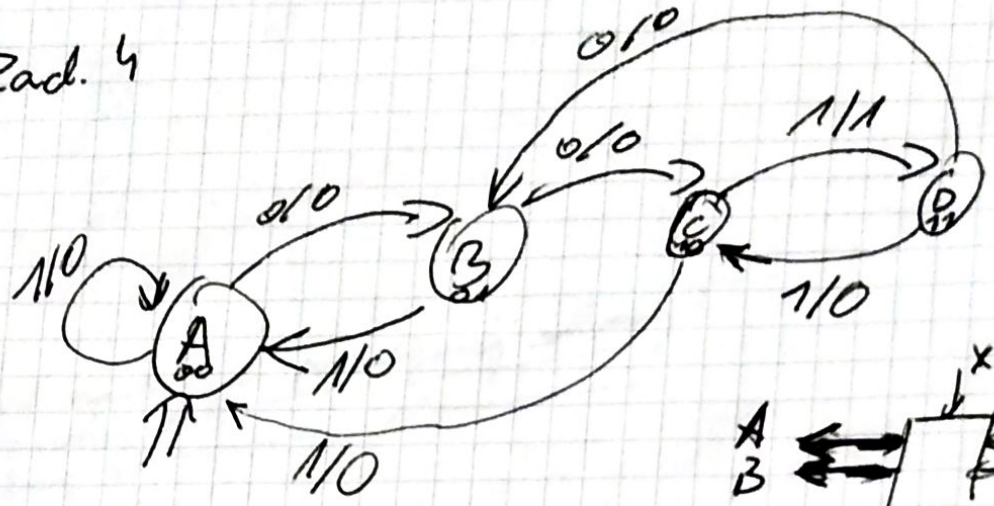


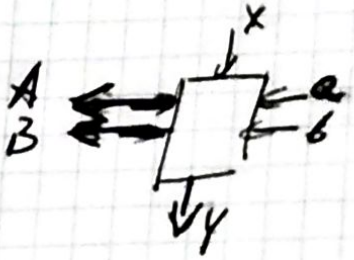


$Y = a + \bar{b}c = \overline{\bar{a} \cdot \overline{\bar{b}c}}$



Zad. 4





A - brak zer
B - nieparzysta liczba zer
C - parzysta liczba zer
D - nieparzysta liczba 1 poprzedzona parzystą liczbą zer

A - 00
B - 01
C - 10
D - 11

ab	x	AB	Y
00	0	01	0
00	1	00	0
01	0	10	0
01	1	00	0
10	0	00	0
10	1	11	1
11	0	01	0
11	1	10	0

a \ bx	00	01	11	10
0	0	0	0	(1)
1	0	(1)	1	0

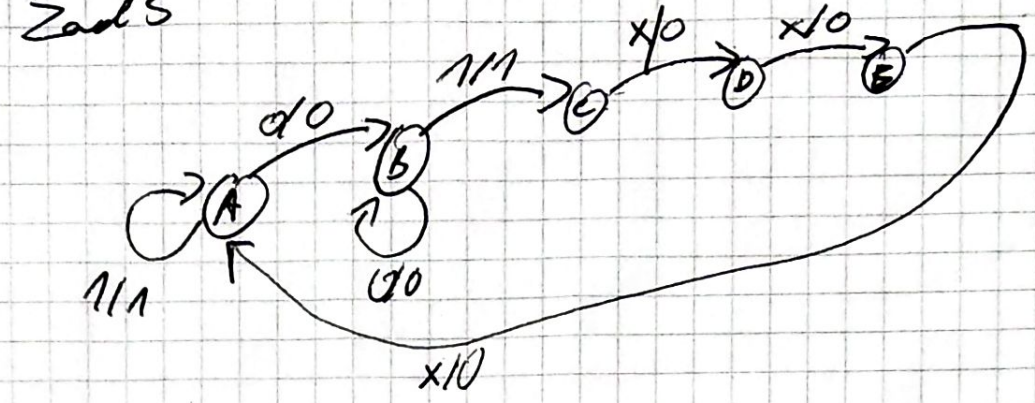
$A = ax + \bar{a}b\bar{x}$

a \ bx	00	01	11	10
0	(1)	0	0	0
1	0	(1)	0	(1)

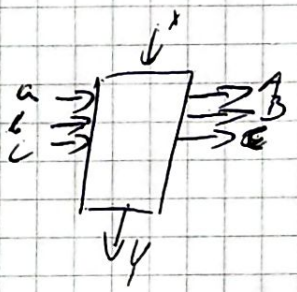
$B = \bar{a}\bar{b}\bar{x} + a\bar{b}x + a\bar{b}\bar{x}$

$Y = a\bar{b}x$

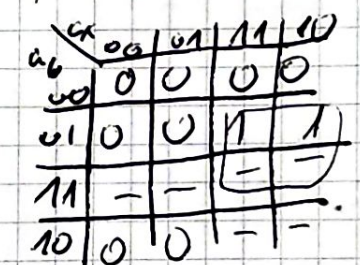
Zad 5



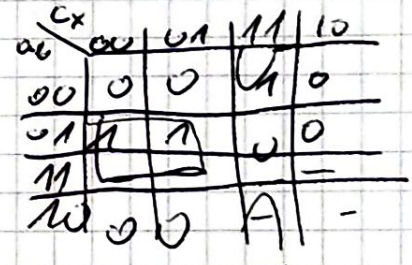
abc
000 A - brak symbolu z poszukiwanej sekwencji
001 B - pierwszy bit
010 C - drugi
011 D - trzeci K-2
100 E - K-3



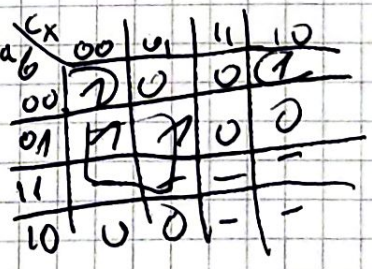
abcx	ABCY
0000	001 0
0001	000 1
0010	001 0
0011	010 1
0100	011 0
0101	011 0
0110	100 0
0111	100 0
1000	000 0
1001	000 0
1010	
1011	
1100	
1101	
1110	
1111	



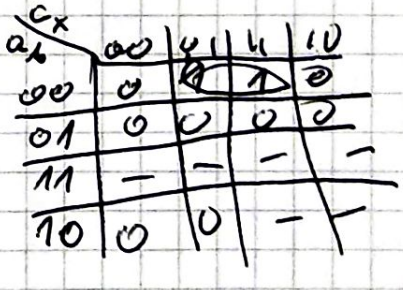
$A = bc$



$B = b\bar{c} + \bar{b}cx$



$C = b\bar{c} + \bar{a}\bar{b}x$



$Y = \bar{a}\bar{b}x$